

Paper: Lang-et al.-Make the most out of your SIMD investments: counter control flow divergence in compiled query pipelines-vldbj2020

Overview of the Main Idea

This paper looks at how modern databases use special hardware instructions called SIMD to handle many data items at once, making queries run much faster. The authors saw that when data is uneven or needs different paths through the query, some parts of the hardware sit idle, this problem is called control flow divergence. They created smart algorithms that keep all parts of the hardware busy, even with tricky, uneven data, so queries don't slow down.

Key Findings/Takeaways

Balancing work between all SIMD lanes (the units inside the CPU that work in parallel) can speed up queries by 25–34% for important types like table scans and joins. Their method works especially well when the data is not uniform (has lots of exceptions or skips), and helps keep query pipelines running at top speed. Unlike older fixes that made pipelines much more complicated, their solution gives a better balance without slowing down later steps.

Description of the System & Modifications

The authors changed how compiled query pipelines assign tasks to SIMD lanes, using new “lane refill” and “push-down parallelism” strategies to make sure no lane is left empty, and modified the DogQC compiler to support this.

Workloads/Benchmarks Used

They tested their strategies on three popular analytic workloads—a table scan (from TPC-H Query 1), a hash join, and a geospatial join showing speed-ups of up to 34% when handling tough, uneven data.